DATA 222 - MODEL

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```
library(car)
```

```
Loading required package: carData
```

```
library(randomForest)
```

```
Warning: package 'randomForest' was built under R version 4.5.3
```

```
randomForest 4.7-1.2
```

```
Type rfNews() to see new features/changes/bug fixes.
```

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.0     v readr     2.1.5
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.1.0
```

```
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::combine()  masks randomForest::combine()
x dplyr::filter()   masks stats::filter()
x dplyr::lag()      masks stats::lag()
x ggplot2::margin() masks randomForest::margin()
x dplyr::recode()   masks car::recode()
x purrr::some()     masks car::some()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
apartments <- read_csv("C:/Users/ritik/OneDrive/Desktop/s8/data222/Final Project/RentWIP_clea
```

```
Rows: 10000 Columns: 8
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): amenities, pets_allowed, cityname, state
dbl (4): bathrooms, bedrooms, square_feet, monthly_price

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
head(apartments)
```

```
# A tibble: 6 x 8
  amenities           bathrooms bedrooms pets_allowed square_feet cityname state
  <chr>                   <dbl>    <dbl> <chr>              <dbl> <chr>    <chr>
1 <NA>                        0        0 none                 101 washing~ dc
2 <NA>                        0        1 none                 106 evansvi~ in
3 <NA>                        1        0 none                 107 arlingt~ va
4 <NA>                        1        0 none                 116 seattle  wa
5 <NA>                        0        0 none                 125 arlingt~ va
6 dishwasher,elevato~         1        0 none                 130 manhatt~ ny
# i 1 more variable: monthly_price <dbl>
```

1. Test-train split

```
set.seed(123)
train <- apartments %>% sample_frac(.70)
test <- anti_join(apartments, train)
```

```
Joining with `by = join_by(amenities, bathrooms, bedrooms, pets_allowed,
square_feet, cityname, state, monthly_price)`
```

2. Linear Regression Model with STATE

```
lin_model<- lm(monthly_price ~ state, data = train)
summary(lin_model)
```

```
Call:
lm(formula = monthly_price ~ state, data = train)

Residuals:
   Min     1Q Median     3Q    Max
 -1996   -358   -117    206  49704

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  981.400    209.420   4.686 2.84e-06 ***
stateal      246.870    259.929   0.950 0.342268
statear       71.075    256.487   0.277 0.781704
stateaz      129.984    232.500   0.559 0.576132
stateca     1815.069    212.608   8.537  < 2e-16 ***
stateco      374.883    218.223   1.718 0.085862 .
statect      330.364    236.726   1.396 0.162894
statedc     1044.531    242.858   4.301 1.72e-05 ***
statede      205.267    579.859   0.354 0.723354
statefl      539.171    217.972   2.474 0.013400 *
statega      293.340    217.267   1.350 0.177015
statehi     2417.378    375.921   6.431 1.36e-10 ***
stateia      -33.593    226.336  -0.148 0.882014
stateid      345.017    341.982   1.009 0.313070
stateil      408.821    218.650   1.870 0.061561 .
statein      172.362    221.261   0.779 0.436009
stateks        4.542    246.424   0.018 0.985294
stateky       14.230    276.303   0.052 0.958929
statela       48.647    253.487   0.192 0.847819
statema     1335.634    226.756   5.890 4.04e-09 ***
statemd      622.633    216.875   2.871 0.004105 **
stateme      881.100    694.569   1.269 0.204642
statemi      305.551    225.805   1.353 0.176048
statemn      377.575    221.971   1.701 0.088986 .
statemo       22.190    220.872   0.100 0.919978
statems      -16.400    411.293  -0.040 0.968195
statemt      774.314    411.293   1.883 0.059792 .
statenc      250.114    216.091   1.157 0.247131
statend      -59.294    232.758  -0.255 0.798927
statene      -59.351    233.850  -0.254 0.799659
statenh      383.156    251.692   1.522 0.127975
statenj      993.371    216.797   4.582 4.69e-06 ***
statenm       -5.733    435.943  -0.013 0.989507
```

```
statenull    433.071    247.094   1.753 0.079706 .
statenv      281.884    233.850   1.205 0.228088
stateny      698.339    250.849   2.784 0.005385 **
stateoh      114.013    218.610   0.522 0.602011
stateok       82.383    226.200   0.364 0.715715
stateor      579.510    223.491   2.593 0.009534 **
statepa      312.220    225.070   1.387 0.165421
stateri      417.600    411.293   1.015 0.309982
statesc      332.804    245.153   1.358 0.174656
statesd      -67.164    244.550  -0.275 0.783600
statetn      356.898    243.404   1.466 0.142617
statetx      178.029    211.147   0.843 0.399173
stateut      288.433    241.818   1.193 0.233000
stateva      517.883    224.083   2.311 0.020855 *
statevt      407.886    326.358   1.250 0.211411
statewa      792.986    215.081   3.687 0.000229 ***
statewi      304.686    219.166   1.390 0.164511
statewv      275.100    694.569   0.396 0.692063
statewy     -268.400    959.685  -0.280 0.779735
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 936.6 on 6948 degrees of freedom
Multiple R-squared:  0.2314,    Adjusted R-squared:  0.2258
F-statistic: 41.02 on 51 and 6948 DF,  p-value: < 2.2e-16
```

This linear model computes the average price of items in various states within the United States. Here, the constant value, 994.241, is the mean price for the base or reference state, and the value of state gives the difference between the price of the reference state and the state under consideration. From the above table, one can conclude that stateCA = 1864.864 implies that California is $1865 higher than the reference state, while stateWV=-199.241 implies that West Virginia is about $199 lower. Based on the small p-values (<0.05), the significant variables include CA, DC, FL, HI, MA, NJ, NY, and WA because they are the primary states within the United States. On average, predictions could be inaccurate by $994. The high R-squared value of 0.2297 implies that 23% of the variance in the price is due to the state variable, hence other causes.

3. Linear Regression Model with square_feet, bedrooms, bathrooms

```r
lin_model2<- lm(monthly_price ~ square_feet + bedrooms + bathrooms, data = train)
summary(lin_model2)
```

```
Call:
lm(formula = monthly_price ~ square_feet + bedrooms + bathrooms,
    data = train)

Residuals:
   Min     1Q Median     3Q    Max
 -6684   -450   -186    215  51047

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 584.27222   29.13769  20.052   <2e-16 ***
square_feet   0.27263    0.02097  12.998   <2e-16 ***
bedrooms    -15.52962   18.01560  -0.862    0.389
bathrooms   482.15225   28.40421  16.975   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 977.6 on 6991 degrees of freedom
  (5 observations deleted due to missingness)
Multiple R-squared:  0.1572,    Adjusted R-squared:  0.1569
F-statistic: 434.8 on 3 and 6991 DF,  p-value: < 2.2e-16
```

The second multiple linear regression model takes square_feet, bedrooms, and bathrooms as independent variables and price as the dependent variable. It is statistically significant ($p <$ 2.2e-16). One extra square foot is associated with an increase in price by about \$0.91. Therefore, the bigger house, the higher its price. In addition, one extra bathroom will increase price by \$273.52. Hence, bathrooms positively impact property prices. A negative coefficient for bedrooms (-\$166.30) implies that the more bedrooms in property with equal square feet and bathrooms, the smaller rooms they have, resulting in reduced property prices. Interception (\$542.42) represents a predicted price when all predictors are zeros. However, it is not a realistic prediction. Furthermore, the R-squared of 0.2021 indicates that predictors explain 20.21% of the total variability of price. Consequently, it is important to consider other aspects like property condition or area that significantly impact price along with the presented predictors.

3. Testing with the linear model

```r
linear_test <- predict(lin_model2, newdata = test)
actual <- test$monthly_price
```

```r
pred <- linear_test

rmse <- sqrt(mean((actual - pred)^2, na.rm = TRUE))
mae <- mean(abs(actual - pred), na.rm = TRUE)

rmse
```

```
[1] 953.771
```

```r
mae
```

```
[1] 536.5369
```

The above are the values of the measures used to evaluate the performance of the second model: RMSE = 799.56 and MAE = 515.99. This implies that the model's estimates are inaccurate by approximately $516 on average, while the average error estimate will be about $800 if higher penalties are put on larger errors.

4. Random Forest Model

```r
train_clean <- na.omit(train)
rf_model <- randomForest(monthly_price ~ ., data = train_clean, ntree = 500)
rf_model
```

```
Call:
 randomForest(formula = monthly_price ~ ., data = train_clean,      ntree = 500)
               Type of random forest: regression
                     Number of trees: 500
No. of variables tried at each split: 2

          Mean of squared residuals: 812111.9
                    % Var explained: 31.58
```

Since the Random Forest model included all the predictors from train_clean, it successfully predicted the variable "monthly_price". The type of the response variable is numeric, thus, this model is called a regression model, while not classification. The number of trees grown is 500; this provides more stability and reliability. At each split of the tree, only two randomly

chosen predictors were considered. Squared residuals' mean (811328.5) indicates the average squared error of prediction, and smaller values correspond to higher accuracy. In addition, 31.65% of variance in the monthly rent was accounted for by this model, indicating that some information about this variable can be found using predictors, while another share of variance remains dependent on additional factors.

5. Testing with Random Forest

```
test_clean <- na.omit(test)
rf_test <- predict(rf_model, newdata = test_clean)
actual_rf <- test_clean$monthly_price
pred_rf <- rf_test

rmse_rf <- sqrt(mean((actual_rf - pred_rf)^2))
mae_rf <- mean(abs(actual_rf - pred_rf))

rmse_rf
```

```
[1] 700.4765
```

```
mae_rf
```

```
[1] 399.567
```

The output of this model gave the RMSE value as 700.48, meaning that the prediction made for monthly rent differs by $700 on average. An MAE value of 399.57 indicates an average error of about $400 per month. It can be inferred that this model provides reasonable predictions with typical prediction errors ranging from hundreds of dollars.

CART modl

```
library(rpart)
library(rpart.plot)

cart_model <- rpart(
  monthly_price ~ bathrooms+bedrooms+pets_allowed+square_feet,
  data = train_clean
```

```
)

cart_model
```

```
n= 4488

node), split, n, deviance, yval
      * denotes terminal node

 1) root 4488 5326985000 1404.245
   2) square_feet< 1102 3590 1322991000 1271.483 *
   3) square_feet>=1102 898 3687751000 1934.999
     6) bedrooms>=1.5 873 1095906000 1873.317
      12) bathrooms< 2.75 809  731004100 1789.449 *
      13) bathrooms>=2.75 64  287280500 2933.469 *
     7) bedrooms< 1.5 25 2472539000 4088.920
      14) square_feet< 1302.5 18   20388410 2042.111 *
      15) square_feet>=1302.5 7 2182830000 9352.143 *
```

```
cart_pred <- predict(cart_model, newdata = test_clean)

cart_rmse <- sqrt(mean((test_clean$monthly_price - cart_pred)^2))
cart_mae <- mean(abs(test_clean$monthly_price - cart_pred))

cart_rmse
```

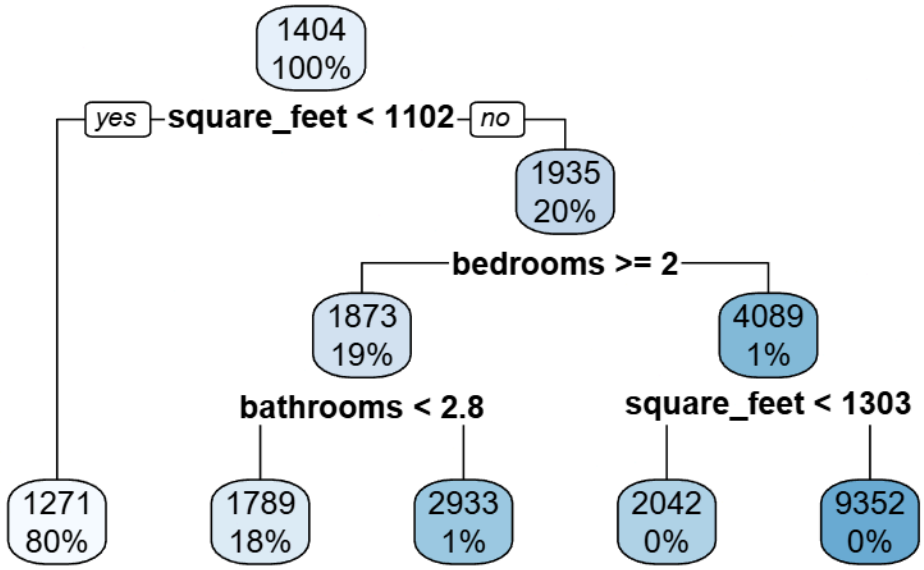
```
[1] 837.7566
```

```
cart_mae
```

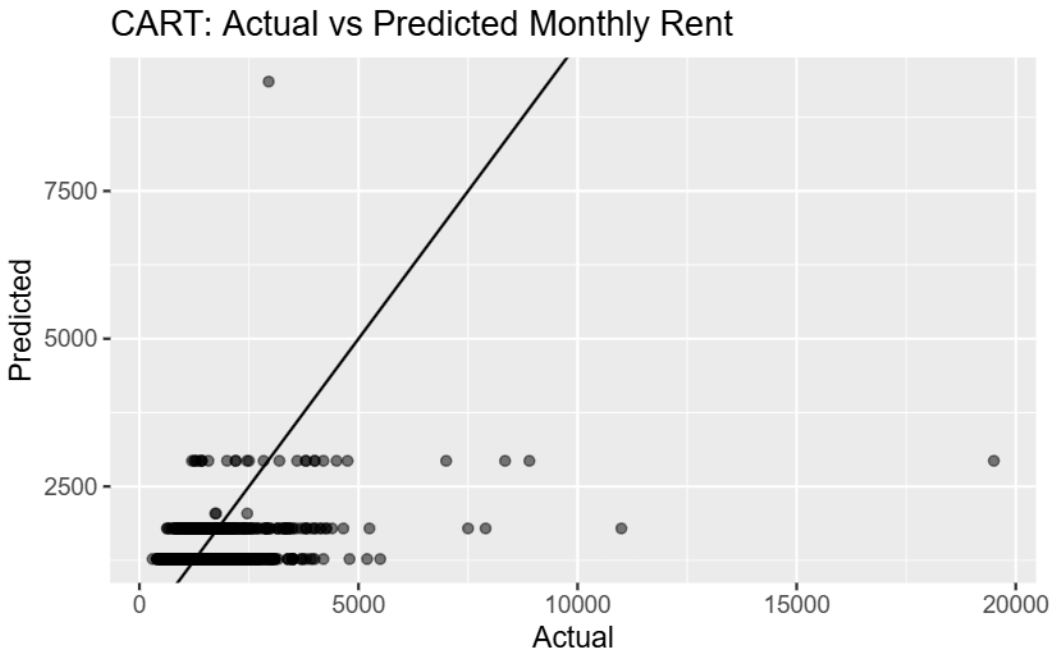
```
[1] 493.8618
```

According to the CART model, the RMSE value is equal to 837.76, meaning that there is an average difference of around $838 between the forecasted monthly rents and the real values considering that large discrepancies carry more significance. An MAE value of 493.86 indicates that the average error is around $494 per month. These relatively high error values indicate that the CART model is less precise than the Random Forest model; however, it can still be used for its interpretable nature and identification of significant variables.

```r
# Tree Plot
rpart.plot(cart_model)
```




```r
# Actual vs Predicted Plot
ggplot(data.frame(
  Actual = test_clean$monthly_price,
  Predicted = cart_pred
), aes(x = Actual, y = Predicted)) +
  geom_point(alpha = 0.5) +
  geom_abline() +
  labs(
    title = "CART: Actual vs Predicted Monthly Rent"
  )
```



Conclusion: The performance of the Random Forest model exceeded that of the CART model due to the smaller amount of error, indicating greater accuracy of predicting the monthly rent prices. It should be expected that the former model would perform better because the Random Forest is based on multiple decision trees, making it capable of finding more complicated relationships, whereas the CART algorithm makes use of one decision tree only. Nevertheless, the latter algorithm was also beneficial for analysis because of the greater interpretability it provided, allowing us to understand the impact of certain features, such as square footage, number of bathrooms and bedrooms, as well as pets allowed. Some features like amenities or city name were excluded from the CART model due to many unique levels.